

- 6 (a)** Neurones in the brain metabolise glucose to produce ATP. Glucose enters brain neurones through transport proteins in the cell surface membrane. These proteins are similar to the transport proteins for glucose found in liver and muscle cells.

State the process by which glucose passes through transport proteins into brain neurones.

..... [1]

- (b)** Brain neurones use ATP very quickly.

One brain neurone uses 4.7×10^9 molecules of ATP per second.

One striated muscle cell uses 9.8×10^6 molecules of ATP per second.

Calculate how many muscle cells would use the same number of ATP molecules per second as one brain neurone.

Give your answer to the nearest whole number.

number of muscle cells = [2]

- (c)** Suggest reasons why neurones, such as those in the brain, use large numbers of ATP molecules per second.

[4]



(d) When ATP is completely broken down, the compound adenosine is released which then leaves the cell.

- During the day, the extracellular concentration of adenosine increases.
- In brain synapses, adenosine binds to adenosine receptors in the cell surface membrane of postsynaptic brain neurones and inhibits postsynaptic neurone activity.
- This action promotes sleep.
- During sleep, adenosine detaches from its receptor and is recycled to form ATP.
- This action promotes wakefulness (the person is awake).

Caffeine is a compound found naturally in tea and coffee and is added to some drinks. Caffeine has a similar shape to adenosine.

Suggest **and** explain the effect on the sleep pattern of a person who has had a drink containing a lot of caffeine.

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[Total: 10]